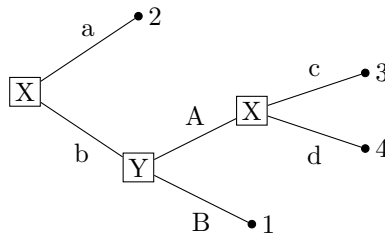


GSOE9210 Engineering Decisions

Problem Set 09

- Consider the following game tree for a zero-sum game (payoffs shown are for player X):



- What are the possible strategies for players X and Y?
- Draw the corresponding normal (matrix) form from the perspective of player X
- Simplify the game by eliminating strategies which aren't best responses
- Does this game have a saddle point?

Solution: Interpret the code 'b;A/c' as follows: "do c at the node arrived at by doing b followed by A". Note that the implied grouping is (b;A)/c. However, because all action names are unique in this example the interpretation b;(A/c) would also be appropriate.

More generally, α/A means "do A at the decision point reached by path α ".

- For X: a, b;A/c, b;A/d For Y: b/A, b/B
- The game matrix is shown below:

		Y		
		b/A	b/B	
X	a	2	2	2
	b;A/c	3	1	1
	b;A/d	4	1	1
		4	2	

		Y		
		b/A	b/B	
X	a	2	2	2
	b;A/c	3	1	1
	b;A/d	4	1	1
		4	2	

- See above right, for the simplification via iterated dominance. The solution after simplification is (a,b/B).
- From the tables above one can see that the row player's *Maximin* value is equal to the column player's *miniMax* value, both being 2. This means there's a saddle point at (a, b/B). Notice that the

saddle point's outcome is the only one which survives elimination of dominated strategies.

2. Find all saddle points for the zero-sum game with the following matrix:

	b_1	b_2	b_3	b_4
a_1	6	-1	1	0
a_2	5	1	7	-2
a_3	3	2	4	3
a_4	-1	0	0	6

Solution: Evaluating the *Maximin* strategies of the two players (recall that the column player is effectively playing *miniMax*):

	b_1	b_2	b_3	b_4	
a_1	6	-1	1	0	-1
a_2	5	1	7	-2	-2
a_3	3	2^*	4	3	2
a_4	-1	0	0	6	-1
	6	2	7	6	

3. Find all saddle points for the following zero-sum game:

	b_1	b_2	b_3	b_4
a_1	4	2	5	2
a_2	2	1	-1	-2
a_3	3	2	4	2
a_4	-1	0	6	1

Solution: Evaluating the *Maximin* strategies of the two players (recall that the column player is effectively playing *miniMax*):

	b_1	b_2	b_3	b_4	
a_1	4	2^*	5	2^*	2
a_2	2	1	1	2	-2
a_3	3	2^*	4	2^*	2
a_4	-1	0	6	1	-1
	4	2	6	2	

	a_1	a_2	a_3	a_4	
a_1	4	2^*	5	2^*	2
a_2	2	1	1	2	-2
a_3	3	2^*	4	2^*	2
a_4	-1	0	6	1	-1
	4	2	6	2	

Here there are four saddle points. Note that each has the same value (2). In general, all saddle points of a zero-sum game must have the same value.

The left matrix shows the simplification steps when using weak dominance, which is the default assumption we use in this course. Notice that already after the first simplification step some of the saddle points (i.e., equilibrium plays) are part of a (weakly) dominated strategy.

In general, if, in the process of simplification, strategies are eliminated using weak dominance, then some equilibria may be eliminated. However, at least one of the saddle points will always survive simplification.

The matrix on the right has been simplified by eliminating only based on strict dominance. Notice that exactly the four equilibria survive.

4. Show that for any zero-sum game in normal (strategic) form the column *miniMax* value is no less than the row *Maximin*.

Consequently, verify that the column maxima are all greater than the row minima.

Solution: Consider a $k \times n$ matrix (i.e., k rows and n columns):

	c_1	$\dots$	c_j	$\dots$	c_q	$\dots$	c_n	m
r_1	$\dots$	$\dots$	a_{1j}	$\dots$	a_{1q}	$\dots$	$\dots$	m_1
$\vdots$			$\vdots$					$\vdots$
r_i	a_{i1}	$\dots$	a_{ij}	$\dots$	a_{iq}^*	$\dots$	a_{in}	m_i^*
$\vdots$			$\vdots$					$\vdots$
r_k	$\dots$	$\dots$	a_{kj}	$\dots$	a_{kq}	$\dots$	$\dots$	m_k
	M_1	$\dots$	M_j^*	$\dots$	M_q	$\dots$	M_n	

Define $m_i = \min_j \{a_{ij}\}$ and $M_j = \max_i \{a_{ij}\}$; i.e., m_i is the minimum value of row i . Similarly, M_j is the maximum of column j .

It follows that $m_i \leq a_{ij}$ for each $1 \leq j \leq n$, and $M_j \geq a_{ij}$ for each $1 \leq i \leq k$.

Define the row player's *Maximin* value as $m^* = \max_i \{m_i\}$, and the column player's *miniMax* value as $M^* = \min_j \{M_j\}$.

We want to show that $m^* \leq M^*$.

Assume that $m^* = m_i$, for some i . Moreover, assume that $m_i = a_{iq}$.

Assume that $M^* = M_j$, for some j . Now $M^* = M_j \geq a_{ij} \geq a_{iq} = m^*$. This concludes the proof.

Moreover, it follows, for any M_j and m_i , that $M_j \geq M^* \geq m^* > m_i$. That is, none of the column player's *miniMax* values are less than any of the row player's *Maximin* values.

This result shows that in a zero-sum game, by playing a *Maximin* strategy neither player can guarantee winning more than what the other player can guarantee losing.

5. Consider the game matrix below for a non zero-sum game in which the row player is R and the column player is C.

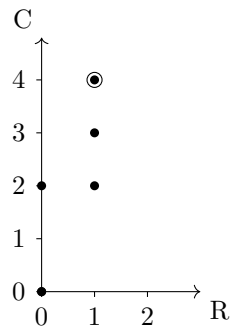
	b_1	b_2	b_3
a_1	0, 0	1, 2	0, 2
a_2	1, 3	1, 4	0, 0

- Identify the equilibrium plays.
- Which plays survive simplification?
- Which plays are Pareto optimal?
- Summarise your findings about non-zero-sum games and the relationship between:
 - plays obtained by dominance elimination and equilibria
 - equilibria and Pareto optimal plays

Solution:

	b_1	b_2	b_3
a_1	0, 0	1, 2	0, 2
a_2	1, 3	1, 4	0, 0

- From the matrix above, equilibrium plays are: (a_1, b_2) , (a_1, b_3) , (a_2, b_2) .
- Only (a_2, b_2) survives. Note that it is an equilibrium play.
-



The only Pareto optimal play is the equilibrium which survives dominance elimination: (a_2, b_2) . Note that in this case there is a Pareto dominant play; i.e., a play that weakly Pareto dominates all other plays.

- Some non-equivalent equilibria are eliminated by dominance. Not all equilibria are Pareto optimal.

6. Show that in a zero-sum game every outcome is Pareto optimal.

Solution: Geometrically, all outcomes in a zero-sum game lie on the off-diagonal; i.e., the line $y = -x$. Intuitively, this shows that no outcome can lie above another in the Pareto sense.

Formally, consider any two arbitrary outcomes in a game with payoffs $(u, -u)$ and $(v, -v)$.

If the two outcomes are equivalent (i.e., $u = v$) the result is trivial.

Consider any two non-equivalent (i.e., distinct) outcomes of a zero-sum game: e.g., $(u, -u)$ and $(v, -v)$, where $u \neq v$. Assume the latter 'Pareto

dominates' the former, and that $v > u$, making the two distinct. It follows that $-u > -v$. But Pareto dominance would require that $-v \geq -u$; i.e., $u \geq v$, which is impossible, since $v > u$. It follows that no outcome can 'Pareto dominate' another distinct outcome; i.e., all distinct outcomes must be Pareto optimal.

Therefore, combining the two cases above, it follows that *all* outcomes must be Pareto optimal.

7. The game 'Matching Pennies' is a two-player zero-sum game in which each player places a covered coin on a table. Each player chooses the face on the coin but keeps it covered so that the other player doesn't know which face is up. The players uncover their coins simultaneously. The 'matching' player (M) wins if the faces on the coin match, and the 'opposites' player (O) wins if the faces are opposite. The winner takes both coins.
 - (a) Represent this game in normal form, showing M's payoffs.
 - (b) Reduce the game using dominance.
 - (c) Find the equilibria of this game.
 - (d) If M believes that O is twice as likely to play heads than tails, what would be his best response?
 - (e) What should be M's best response if he believes O will play heads and tails with equal likelihood?
 - (f) Repeat question 7e above for O.
 - (g) If the game is repeated many times, what strategy should the players play?

Solution:

- (a) The game matrix is shown below:

		O	
		h	t
M	H	1	-1
	T	-1	1

- (b) Neither strategy, for either player, is dominated.
- (c) The game has no equilibrium plays/points among the pure strategies, however see below.
- (d) Suppose M's belief about O is represented by the probability distribution over O's strategies $\beta \sim \frac{2}{3}h\frac{1}{3}t$. Because belief is modelled by a probability distribution, we can use *Bayes* values to evaluate actions. The *Bayes* values for each of M's strategies are:

$$\begin{aligned}
 V_B^\beta(H) &= \frac{2}{3}(1) + \frac{1}{3}(-1) \\
 &= \frac{2}{3} - \frac{1}{3} = \frac{1}{3} \\
 V_B^\beta(T) &= \frac{2}{3}(-1) + \frac{1}{3}(1) \\
 &= -\frac{2}{3} + \frac{1}{3} = -\frac{1}{3}
 \end{aligned}$$

The best response is H.

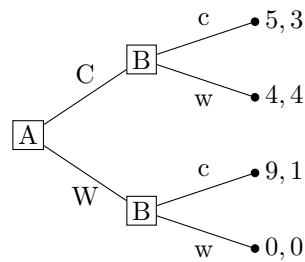
- (e) Suppose M's belief about O is represented by the probability distribution over O's strategies $\beta \sim \frac{1}{2}h\frac{1}{2}t$. The *Bayes* values for each of M's strategies are:

$$\begin{aligned} V_B^\beta(H) &= \frac{1}{2}(1) + \frac{1}{2}(-1) \\ &= \frac{1}{2} - \frac{1}{2} = 0 \\ V_B^\beta(T) &= \frac{1}{2}(-1) + \frac{1}{2}(1) \\ &= -\frac{1}{2} + \frac{1}{2} = 0 \end{aligned}$$

Both H and T are best responses.

- (f) Because the game is symmetric, the results are the same for O.
- (g) If the game is repeated many times then it makes sense for players to also consider mixed strategies. In this case, from the previous question, if O plays the mixture $\frac{1}{2}h\frac{1}{2}t$, then $V_B^\beta(H) = V_B^\beta(T) = 0$. If O were to play any other strategy, M would have a counter-strategy which would increase M's payoff above 0. Similarly, for any mixture $M(\mu) = \mu H + (1 - \mu)T$ of M's strategies, $V_B^\beta(M(\mu)) = 0$. Hence, all of M's mixtures are best responses to O's $\frac{1}{2}h\frac{1}{2}t$. This holds in particular for M's mixture $M(\frac{1}{2}) = \frac{1}{2}H\frac{1}{2}T$. But if M plays $\frac{1}{2}H\frac{1}{2}T$, then, by the symmetry of the game/matrix, any mixed strategy of O's will be a best response to this strategy. In particular, $m(\frac{1}{2}) = \frac{1}{2}h\frac{1}{2}t$, will be a best response for O. It follows that the play $(\frac{1}{2}H\frac{1}{2}T, \frac{1}{2}h\frac{1}{2}t)$, is a 'mixed strategy equilibrium' of the game. In general, by a similar reasoning, for any zero-sum game there exists at least one mixed-strategy equilibrium.

8. The following game tree represent the example in lectures for the case in which Alice the gorilla moves first and Bob the monkey moves second.



- (a) What is Bob's best response to Alice if she waits? If Alice climbs?
- (b) Are there any equilibrium points? If so, which are they?
- (c) Rationalise the game using dominance.

Solution:

- (a) Note that Bob's strategies must specify what to do at both the C-node and the W-node.

From the tree it is evident that Bob's best responses to Alice's W are both $\frac{C}{W/c}$ and $\frac{C}{W/w}$. Note that, in this case (Alice chooses W), the response to C in Bob's strategy is irrelevant.

For Alice's C, Bob's best response involves waiting; i.e., both $\frac{W}{C/w}$ and $\frac{W}{C/c}$ are best responses. In this case, Bob's response to W is irrelevant.

- (b) Alice's best response to either of $\frac{C}{W/c}$ and $\frac{C}{W/w}$ is W. So $\left(W, \frac{W}{C/w}\right)$ and $\left(W, \frac{W}{C/c}\right)$ are both equilibrium plays.

For Bob's $\frac{W}{C/w}$, Alice's best response is W. So of these two only $\left(C, \frac{W}{C/w}\right)$ is an equilibrium play.

Therefore the equilibrium plays are:

$\left(W, \frac{W}{C/w}\right)$, $\left(W, \frac{W}{C/c}\right)$, and $\left(C, \frac{W}{C/w}\right)$, with values (9, 1), (9, 1), (4, 4) respectively.

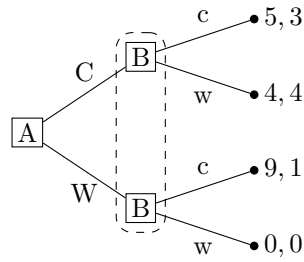
- (c) The only (sub-game perfect) equilibrium is $\left(W, \frac{C}{W/c}\right)$. This can be verified from the game matrix below:

		B			
		$\frac{C}{W/w}$	$\frac{C}{W/c}$	$\frac{C}{C/c}$	$\frac{C}{C/w}$
A	C	4, 4 [*]	4, 4 [*]	5, 3 [*]	5, 3 [*]
	W	0, 0 [*]	9, 1 [*]	0, 0 [*]	9, 1 [*]

Note that because Bob's $\frac{C}{W/c}$ is a sub-optimal response to Alice's C (it is dominated by Bob's $\frac{C}{W/w}$), then $\left(W, \frac{C}{W/c}\right)$ can be ignored as a solution to the game (it is said to not be *sub-game perfect*). A similar argument would eliminate $\left(C, \frac{C}{W/w}\right)$, as Bob should not choose it in preference to $\frac{C}{W/c}$.

This is an example of a non-zero sum game in which (the outcomes of) some equilibria may belong to weakly dominated strategies, and hence be eliminated.

9. If Alice and Bob move simultaneously we have the game tree below.



- Draw the game matrix.
- Are there any equilibrium plays/points? If so, which are they?
- If Bob had an injured arm, indicating that he would be less likely to climb than to wait, which would be the rational solution?

Solution:

- Neither Bob nor Alice have the information to make strategies available to them that allow them to react to each other's move.

	c	w
C	5, 3	4, 4*
W	9, 1*	0, 0

- Equilibrium plays are those in which the players to opposites: (C,w) and (W,c).
- Suppose Alice believed that Bob is less likely to climb than to wait; i.e., $P(c) < P(w)$. Let $p = P(c)$. Then Alice's belief about Bob would be $\beta = (P(c), P(w)) = (p, 1 - p)$, where $p < \frac{1}{2}$. Now:

$$\begin{aligned}
 V_B^\beta(C) &= p(5) + (1 - p)(4) \\
 &= 4 + p \\
 V_B^\beta(W) &= p(9) + (1 - p)(0) \\
 &= 9p
 \end{aligned}$$

It follows that Alice should climb if:

$$\begin{aligned}
 V_B^\beta(C) &\geq V_B^\beta(W) \\
 4 + p &\geq 9p \\
 4 &\geq 8p \\
 p &\leq \frac{1}{2}
 \end{aligned}$$

In this case simplification doesn't eliminate any strategies (i.e., assign them a belief of 0), but we have some information about relative likelihoods; i.e., $P(c) < P(w)$. This implies that $p < 1 - p$, and therefore that $p < \frac{1}{2}$.

Because Alice prefers to climb in response to her belief that Bob is less likely to do so (i.e., given $P(c) < P(w)$), and because Bob knows this, the equilibrium (C,w) is the rational solution.

10. Show that if a zero-sum game has a dominant row and column, then the two determine an equilibrium play.

Solution: A row that dominates must have values that are maximal in each column. Similarly, for a dominant column all values in that column must be minimal in each row. Hence the corresponding pair of strategies must be minimal in their row and maximal in its column. This makes the outcome a saddle point, and hence the play an equilibrium.